Project Proposal: Noise-Adaptive Decomposition of Multi-Qubit Gates for IBM Quantum Devices

Project Lead Details:

Name: Aarav Ratra
Current Affiliation: TU Delft (MSc Quantum Information Science and Technology [Ongoing] 2025-27)
Past Affiliation: IIT Roorkee (BTech Engineering Physics 2021-25)

Note: A co-mentor would also be welcome for this project, as the topic is interdisciplinary and the project is currently self-directed.

Objectives

- Study how Qiskit's transpiler handles decomposition of multi-qubit gates for IBM backends and how device noise information can be incorporated into compilation decisions.
- Build a catalogue of algebraically equivalent decompositions for selected gates in the native basis of IBM devices.
- Define and test a noise-aware cost function for selecting decompositions using backend calibration data.
- Integrate the resulting strategy into Qiskit's transpilation workflow and benchmark it against the default pipeline.

Project Idea and Motivation

This project aims to implement and evaluate a noise-adaptive decomposition strategy for Qiskit's transpiler. For a given IBM backend, the idea is to use calibration data and native basis gates to choose among multiple algebraically equivalent decompositions of multi-qubit gates such as CCX/Toffoli and CCZ, so that the selected implementation minimizes an error-based cost derived from the backend noise profile.

Current transpilation pipelines use fixed decomposition rules from equivalence libraries and generally do not adapt this choice to device-specific noise characteristics. However, many multi-qubit gates admit several valid decompositions into native 1- and 2-qubit gates, with different depths, gate counts, and error footprints. Making this choice noise-aware may improve output fidelity without requiring any changes to the hardware itself.

Research questions

- For IBM backends, to what extent does noise-adaptive decomposition improve simulated and, where feasible, hardware-evaluated output fidelity compared to Qiskit's default decompositions?
- How do these gains vary across benchmark families such as QFT, Grover, and QAOA using MQT Bench circuits?

Scope

Given the short duration, the project is intentionally scoped to a focused and feasible implementation.

- **Target platform:** IBM fake backends for development and evaluation, with testing on IBM Quantum backends where feasible and if time permits.

- **Gate set focus:** CCX, CCZ, CZ, and SWAP, with optional extension to Fredkin if time permits.
- **Noise model:** Gate error rates extracted from backend calibration data, with device-average costs as the primary starting point.
- **Compilation focus:** Integration at an appropriate early stage of Qiskit's transpilation pipeline.
- **Benchmarking focus:** Standard algorithm-level circuits from MQT Bench, such as QFT, GHZ, Grover, and QAOA.

Work Plan (1st July to 15th August)

- **Weeks 1–2: Transpiler, Noise Models, and Device Basis Sets / Gate Equivalences and Decomposition Catalog**
  Understand Qiskit's transpiler stages, learn how noise models are constructed and extracted from a device, and explore native basis gates. In parallel, study gate equivalences and implement a catalog of alternative decompositions for multi-qubit gates (CCX, CCZ, CZ, SWAP) expressed in the device's native basis.
- **Week 3: Cost Function and Optimal Decomposition Selection**
  Define a cost function using the extracted noise profile, score all candidate decompositions, and select the lowest-error option for each gate type.
- **Week 4: Integration into the Compilation Pipeline**
  Implement a custom Qiskit compilation pass that applies the noise-optimal decompositions and integrate it into the preset transpiler at the appropriate stage.
- **Week 5: Benchmarking and Analysis**
  Use MQT Bench to compare the baseline Qiskit transpiler against the noise-aware pipeline across standard benchmark circuits, evaluating depth, gate count, and output fidelity.
- **Week 6: Documentation and Presentation**
  Consolidate the implementation and results into a final report/presentation and summarize the main findings of the project.

Deliverables

- A Qiskit TransformationPass implementing noise-adaptive decomposition for CCX/CCZ/CZ/SWAP.
- A decomposition catalog in terms of the backends' basis gates.
- Benchmark scripts using MQT Bench + Qiskit Aer noise models.

Intern Profile and Team Structure

- **Preferred team size:** 2 interns. A small team is preferred so that responsibilities can be divided naturally while keeping discussions and mentoring manageable.
- **Expected background:** Interns should be comfortable with basic quantum circuits, standard gate-based computation, Python programming, and Qiskit. They should also be willing to read technical documentation and engage with Qiskit's internal transpilation workflow.
- **Suggested split of responsibilities:** Depending on interest and background, one intern may focus more on the mathematical side of the project, such as gate equivalences, decomposition structures, and cost-function reasoning, while another may focus more on implementation, transpiler integration, and benchmarking.
- **Working style:** This project is intended to be collaborative and exploratory. While the broad direction is defined, there is room for interns to shape parts of the work through their own reading, implementation choices, and proposed extensions. Initiative, curiosity, and comfort with open-ended technical work will therefore be especially valuable.